

CSCE 606 – Software Engineering
Project: Inventory System
Sprint # 4 – Retrospective

Customer: Dr. Robert Lightfoot

Links:

- Deployed app: <https://robert-inventory-f91e26f91bb2.herokuapp.com/>
- Github Project: <https://github.com/orgs/tamu-edu-students/projects/101>
- Github Repo: <https://github.com/tamu-edu-students/CSCE-606-Inventory-System>
- Shared folder:
<https://drive.google.com/drive/folders/1NsQfvHf9n272E8VKVghdkU4B7Elk4EYI>
- Discussion Channel: <https://app.slack.com/client/T08A4A8U89M/C08AJRT0UTC>

Spring #4 Dates:

- Started date: 03/15/2025
- Finished date: 04/01/2025

Team and Members contribution:

Name	Role	Contribution (%)	Points	Comments
Rafael Morales	DEV	20	7	Contributed to the improvement of the dashboard response for mobile, fixed several bugs on the JS file for searching. Validated and consolidated models for new Location features. Helped to fix /bins response for mobile devices
Khush Patel	SM/DEV	18	6	Contributed to the improvement of the pop up windows response for mobile. Contributed in the logic of the Log controller.
Saba Mostofi	DEV	22	8	Contributed to the improvement of /bins and /items response for mobile. Introduced a sharing feature where 2 users can share the /bin and /item data.
Arun Akash	PO/DEV	22	8	Contributed to the improvement of

				/locations and /items response to mobile devices.
Sid Rayapur	DEV	18	6	Contributed in the logic for the log controller. Wrote the sprint 4 plan documentation

Sprint #4 Goal:

- Improve User Mobile Experience for Dashboard. ✓
- Improve Items Webpage for a Better User Mobile Experience. ✓
- Improve bins /bins for a better user mobile experience ✓
- Add capabilities on /bins to remove, add, edit items targeting user mobile experience ✓
- Improve pop-up windows for a better mobile user experience ✓
- Create a Garage sale report targeting user mobile experience ✓
- Improve /locations webpage for a better user mobile experience ✓

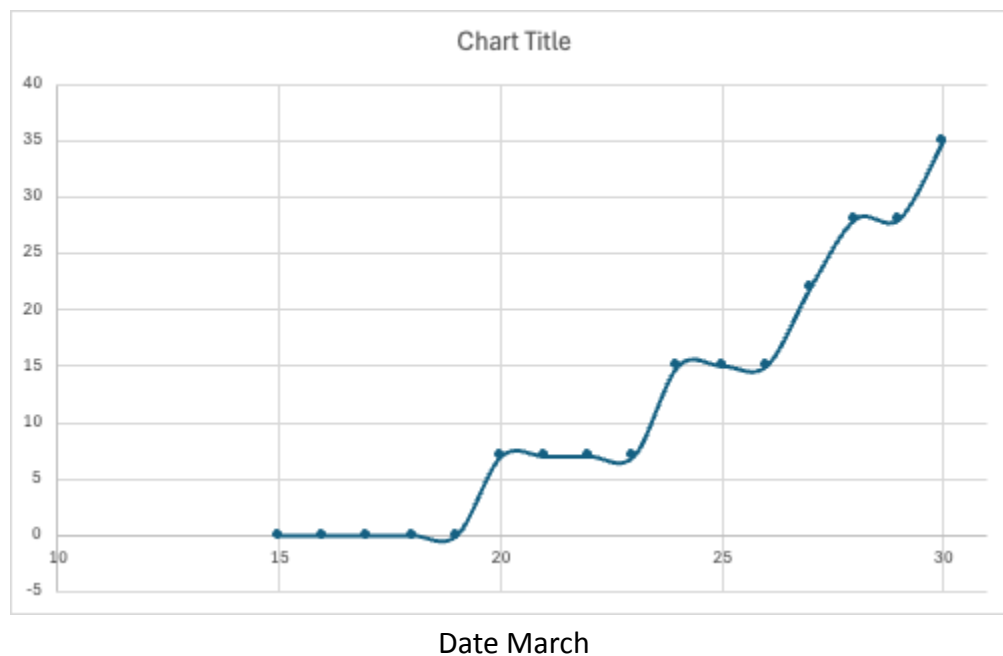
Sprint #4 Achievements:

- Dashboard was improved for mobile devices. Currently, all its features can fit in a mobile display size.
- /items /bins and locations now have a more consistent UI. Every object has its own block container that is more responsive for mobile devices
- Several requests from the last customer meeting were implemented, including the capability to add items or delete items when you are inside a bin webpage.
- Mini pop-up windows were redesigned for a better mobile response.
- SALE links on the navigation bar are finally active. You can generate garage sales and log reports depending on the input dates.
- Several bugs regarding the deletion of /bin and /items were fixed. Now, the User can remove any time from the bin, and permanently delete it once it doesn't belong to any bin.

Sprint #4 Backlog:

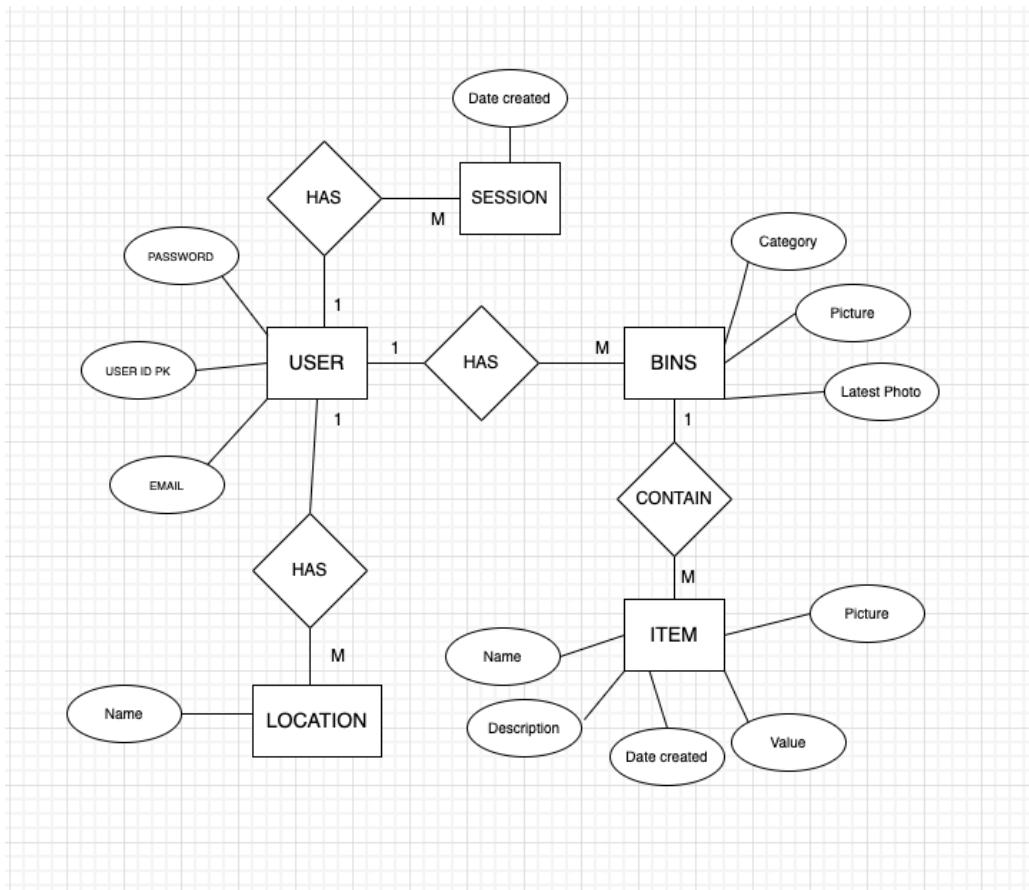
Backlog Item	Status	Notes
Dashboard improvement response for mobile screens	✓ Completed	The implementation was completed.
/bins /items /locations improvement response for mobile screens	✓ Completed	The implementation was completed.
Pop-up windows redesign for mobile screens	✓ Completed	The implementation was completed.
- Create LOG reports	✗ Not Completed	The implementation was more difficult than expected. It will be included in the final product.
Create SALE reports	✓ Completed	The implementation was completed.
+ Add User sharing capabilities	✓ Completed	Last minute request from the customer
+ fix flash messages	✓ Completed	All the message are part of JS

Burn down Chart:



Design Diagrams:

This is the current state of the Database:



Dashboard:

[HOME](#) [BINS](#) [ITEMS](#) [LOG](#) [SALE](#)

A

My Inventory

Search

Search

Search by Location

/bins

HOMEBINSITEMSLOGSALE

A

My Bins

+ Create New Bin

All Categories

Sort by Name

Server Room Equipment

Warehouse A

IT

Items in this bin:

Rack Server

UPS Battery

Thank you

Networking

Tech Lab

IT

Items in this bin:

Cisco Router

Ethernet Switch

Thank you

/items

My Items

+ New Item

Search items by name...

Filter by value...

Reset

Rack Server

Server Room Equipment
Warehouse A
\$2,500.00



UPS Battery

Server Room Equipment
Warehouse A
\$400.00



HOMEBINSITEMSLOGSALE

A

Your Locations

+ New Location

Search locations by name...

Warehouse A

1 Bin

2 Items

Thank you

Tech Lab

1 Bin

2 Items

Thank you

Living Room

0 Bins

0 Items

Documents of change:

Design Change	Comments
LOG, SALE	Log and Sale links on the Navigation Tab are now active
Bins and Items Webpage	Bins Items and Location were completely redesigned to improved the Mobile User experience

Customer Meeting:

- date: 03/21/2025 Via Zoom.

Evaluation of Code and Test Quality

All Files (94.46%)	Controllers (94.08%)	Channels (100.0%)	Models (97.26%)	Mails (100.0%)	Helpers (100.0%)	Jobs (0.0%)	Libraries (100.0%)
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All Files (94.46% covered at 12.63 hits/line)

26 files in total.

379 relevant lines, 358 lines covered and 21 lines missed. (94.46%)

File	% covered ▲	Lines	Releva
🔍 app/jobs/application_job.rb	0.00 %	7	
🔍 app/controllers/dashboard_controller.rb	83.33 %	12	
🔍 app/controllers/items_controller.rb	90.00 %	155	
🔍 app/controllers/locations_controller.rb	91.67 %	67	
🔍 app/controllers/users_controller.rb	91.67 %	21	
🔍 app/models/item.rb	92.00 %	46	
🔍 app/controllers/bins_controller.rb	94.92 %	120	
🔍 app/controllers/password_controller.rb	95.92 %	97	
🔍 app/controllers/application_controller.rb	100.00 %	45	
🔍 app/controllers/search_controller.rb	100.00 %	18	

Cucumber Scenarios and Results:

- Items modifications


```

Background:                                     # features/sprint4/items_index.feature:3
  Given I am a logged in user                   # features/step_definitions/items_index_steps.rb:1
  And there are some items created for me        # features/step_definitions/items_index_steps.rb:6

Scenario: User sees the items index page with header and buttons # features/sprint4/items_index.feature:7
Capybara starting Puma...
* Version 6.6.0, codename: Return to Forever
* Min threads: 0, max threads: 4
* Listening on http://127.0.0.1:63632
  When I visit the items-page                   # features/step_definitions/items_index_steps.rb:18
  Then I should see the caption "My Items"      # features/step_definitions/items_index_steps.rb:22
  And I should see a "New Item" button          # features/step_definitions/items_index_steps.rb:26
  And I should see a "Back to Dashboard" button # features/step_definitions/items_index_steps.rb:26

Scenario: User sees the filter inputs            # features/sprint4/items_index.feature:13
  When I visit the items-page                   # features/step_definitions/items_index_steps.rb:18
  Then I should see a text input with id "filterInput" # features/step_definitions/items_index_steps.rb:34
  And I should see a number input with id "valueFilter" # features/step_definitions/items_index_steps.rb:38
  And I should see a button with id "resetFilters" # features/step_definitions/items_index_steps.rb:42

@javascript
Scenario: User sees a list of item cards        # features/sprint4/items_index.feature:20
  When I visit the items-page                   # features/step_definitions/items_index_steps.rb:18
  Then I should see item cards for all my items # features/step_definitions/items_index_steps.rb:46

@javascript
Scenario: User filters items by name            # features/sprint4/items_index.feature:25
  When I visit the items-page                   # features/step_definitions/items_index_steps.rb:18
  When I filter items by name "screwdriver"      # features/step_definitions/items_index_steps.rb:50
  Then I should only see item cards containing "screwdriver" # features/step_definitions/items_index_steps.rb:61

@javascript
Scenario: User filters items by value          # features/sprint4/items_index.feature:31
  When I visit the items-page                   # features/step_definitions/items_index_steps.rb:18
  When I filter items by value "50"             # features/step_definitions/items_index_steps.rb:56
  Then I should only see item cards with value less than or equal to 50 # features/step_definitions/items_index_steps.rb:67

```

```

@javascript
Scenario: User filters items by value          # features/sprint4/items_index.feature:31
  When I visit the items-page                   # features/step_definitions/items_index_steps.rb:18
  When I filter items by value "50"             # features/step_definitions/items_index_steps.rb:56
  Then I should only see item cards with value less than or equal to 50 # features/step_definitions/items_index_steps.rb:67

@javascript
Scenario: User sees no matching results        # features/sprint4/items_index.feature:37
  When I visit the items-page                   # features/step_definitions/items_index_steps.rb:18
  When I filter items by name "doesnotexist"    # features/step_definitions/items_index_steps.rb:50
  Then I should see the message "No items to display" # features/step_definitions/items_index_steps.rb:73

Scenario: User sees no items message when no items exist # features/sprint4/items_index.feature:42
  Given I have no items                         # features/step_definitions/items_index_steps.rb:14
  When I visit the items-page                   # features/step_definitions/items_index_steps.rb:18
  Then I should see the message "No items yet"  # features/step_definitions/items_index_steps.rb:73
  And I should see a button labeled "Add Item"  # features/step_definitions/items_index_steps.rb:77

7 scenarios (7 passed)
37 steps (37 passed)
0m6.783s

```

- Location changes:

```
* Version 6.6.0, codename: Return to Forever
* Min threads: 0, max threads: 4
* Listening on http://127.0.0.1:63659
  When I visit the locations page # features/step_definitions/locations_index_steps.rb:20
  Then I should see the heading "Your Locations" # features/step_definitions/locations_index_steps.rb:24
  And I should see "New Location" button # features/step_definitions/locations_index_steps.rb:28
  And I should see a "Back to Dashboard" button # features/step_definitions/items_index_steps.rb:26

Scenario: User sees the search bar # features/sprint4/locations_index.feature:13
  When I visit the locations page # features/step_definitions/locations_index_steps.rb:20
  Then I should see a text input with id as "locationSearch" # features/step_definitions/locations_index_steps.rb:32

@javascript
Scenario: User sees all location cards # features/sprint4/locations_index.feature:18
  When I visit the locations page # features/step_definitions/locations_index_steps.rb:20
  Then I should see location cards for all my locations # features/step_definitions/locations_index_steps.rb:36

@javascript
Scenario: User filters locations by name # features/sprint4/locations_index.feature:23
  When I visit the locations page # features/step_definitions/locations_index_steps.rb:20
  And I filter locations by name "warehouse" # features/step_definitions/locations_index_steps.rb:40
  Then I should only see location cards containing "warehouse" # features/step_definitions/locations_index_steps.rb:45

@javascript
Scenario: User sees message when no locations match search # features/sprint4/locations_index.feature:29
  When I visit the locations page # features/step_definitions/locations_index_steps.rb:20
  And I filter locations by name "notfound" # features/step_definitions/locations_index_steps.rb:40
  Then I should see the error "No locations to display" # features/step_definitions/locations_index_steps.rb:55

@javascript
Scenario: User sees the add, edit, delete, and warning modals # features/sprint4/locations_index.feature:35
  When I visit the locations page # features/step_definitions/locations_index_steps.rb:20
  Then I should see the modal with id "addLocationModal" # features/step_definitions/locations_index_steps.rb:51
  And I should see the modal with id "editLocationModal" # features/step_definitions/locations_index_steps.rb:51
  And I should see the modal with id "deleteLocationModal" # features/step_definitions/locations_index_steps.rb:51
  And I should see the modal with id "warningModal" # features/step_definitions/locations_index_steps.rb:51
```

```
@javascript
Scenario: User adds a new location # features/sprint4/locations_index.feature:43
  When I visit the locations page # features/step_definitions/locations_index_steps.rb:20
  When I open the Add Location modal # features/step_definitions/locations_index_steps.rb:59
  And I fill in the name "My New Location" # features/step_definitions/locations_index_steps.rb:64
  And I submit the add location form # features/step_definitions/locations_index_steps.rb:70
  Then I should see a location card with name "My New Location" # features/step_definitions/locations_index_steps.rb:76

@javascript
Scenario: User edits a location # features/sprint4/locations_index.feature:51
  When I visit the locations page # features/step_definitions/locations_index_steps.rb:20
  When I open the Edit Location modal for the first location # features/step_definitions/locations_index_steps.rb:80
  And I update the name to "Renamed Location" # features/step_definitions/locations_index_steps.rb:85
  And I submit the edit location form # features/step_definitions/locations_index_steps.rb:91
  Then I should see a location card with name "Renamed Location" # features/step_definitions/locations_index_steps.rb:76

@javascript
Scenario: User deletes a location # features/sprint4/locations_index.feature:59
  When I visit the locations page # features/step_definitions/locations_index_steps.rb:20
  And the first location does not have bins and items # features/step_definitions/locations_index_steps.rb:97
  And I open the Delete Location modal for the first location # features/step_definitions/locations_index_steps.rb:105
  And I confirm the delete action # features/step_definitions/locations_index_steps.rb:110
  Then I should not see the deleted location on the page # features/step_definitions/locations_index_steps.rb:116

@javascript
Scenario: User tries to delete a location with bins/items # features/sprint4/locations_index.feature:67
  When I visit the locations page # features/step_definitions/locations_index_steps.rb:20
  And the first location has bins and items # features/step_definitions/locations_index_steps.rb:105
  And I click delete on the first location # features/step_definitions/locations_index_steps.rb:110
  Then I should see the warning modal # features/step_definitions/locations_index_steps.rb:116
  And I should see "Please delete the bins and the associated items before you delete this location." # features/step_definitions/locations_index_steps.rb:124
```

```
10 scenarios (10 passed)
59 steps (59 passed)
0m4.953s
```

- Updating Bin Location

```
Feature: Update Bin Location
  As a user
  I want each bin to reference a location
  So that items are properly categorized

  Scenario: Assign a location to a bin # features/sprint3/proper_
    Given I am a logged-in user # features/step_definitions/
    And a location "Warehouse A" exists # features/step_definitions/
    When I visit the new bin page # features/step_definitions/
    And I fill in the bin "Location" with "Warehouse A" # features/step_definitions/
    Then the bin should be associated with "Warehouse A" # features/step_definitions/

Feature: Search Bars on Dashboard
  As a user of the website,
  I want to have two separate search bars,
  So that I can quickly find my bins/categories and items.

  Background: # fea
    Given I am a logged-in user on the dashboard # fea
    And I have a valid location # fea
    And I have bins with names "Storage Bin" and "Office Supplies" # fea
    And I have items with names "drill" and "screwdriver" on bin "Storage Bin" # fea
    And I have items with names "staple" and "eraser" on bin "Office Supplies" # fea

  ### ✅ SEARCH BAR FOR BINS/CATEGORIES ##
  @javascript
  Scenario: Searching for a bin by name # features/sprint3/search_bar
    When I enter "Storage Bin" in the bins search bar # features/step_definitions/se
    And I click the bins search button # features/step_definitions/se
    Then I should see bins matching "Storage Bin" # features/step_definitions/se
2025-03-21 09:15:19 WARN Selenium [:clear_session_storage] [DEPRECATION] clear_sessi
2025-03-21 09:15:19 WARN Selenium [:clear_local_storage] [DEPRECATION] clear_local_s
```

Some RSPEC Examples:

- Changes on bin model:

```
(db608) rafael@MacBook-Pro CSCE-606-Inventory-System % bundle exec rspec spec/models/bin_spec.rb
.✅ Test Passed: requires name
.✅ Test Passed: belongs to user
.✅ Test Passed: generated qr code upon creation
.✅ Test Passed: has a valid QR code
.✅ Test Passed: Associations - belong to user
.✅ Test Passed: Association, has many items
.✅ Test Passed: Association: Ensure QR code belongs to correct user
.✅ Test Passed: Instance Method, return item belonging to bin
.✅ Test Passed: Instance Method, return empty when bin has no item
.✅ Test Passed: User association, increment user bins when created
.✅ Test Passed: User association, decrements user bins count when destroyed
.✅ Test Passed: bin is not empty
.✅ Test Passed: maintain association with items
.✅ Test Passed: allows items to be reassigned before deletion
.
Finished in 0.42312 seconds (files took 1.15 seconds to load)
14 examples, 0 failures
```

- Changes on location model:

```
(db608) rafael@MacBook-Pro CSCE-606-Inventory-System % bundle exec rspec spec/models/location_spec.rb
✓ Test Passed: is valid with a name and a user
✓ Test Passed: is invalid without a name
✓ Test Passed: is invalid without a user
.

Finished in 0.04095 seconds (files took 1.12 seconds to load)
3 examples, 0 failures
```

- Changes on /item

```
(db608) rafael@MacBook-Pro CSCE-606-Inventory-System % bundle exec rspec spec/models/item_spec.rb
✓ Test Passed: Validation
✓ Test Passed: Requires Name
✓ Test Passed: Validates value is greater than or equal to 0
✓ Test Passed: Association
✓ Test Passed: Has many attached item pictures
✓ Test Passed: CRUD Save
✓ Test Passed: CRUD Retrieve
✓ Test Passed: CRUD retrieve Storage
✓ Test Passed: CRUD Retrieve items by user
✓ Test Passed: CRUD UPDATE
✓ Test Passed: Moves
✓ Test Passed: CRUD Delete
```

- Changes in dashboard

```
Stopped processing Simplecov as a previous error not related to Simplecov has been detected
(db608) rafael@MacBook-Pro CSCE-606-Inventory-System % bundle exec rspec spec/views/dashboard
✓ Test Passed: view nav links
✓ Test Passed: inputs and buttons
✓ Test Passed: search by location button
✓ Test Passed: display user profile icon
.

Finished in 0.1576 seconds (files took 1.13 seconds to load)
4 examples, 0 failures
```

Note:

Some Cucumber and RSpec scenarios for the new UI For Bins and Items, were already implemented in the previous sprints. We made sure that those scenarios were still passing the test without any issue.